

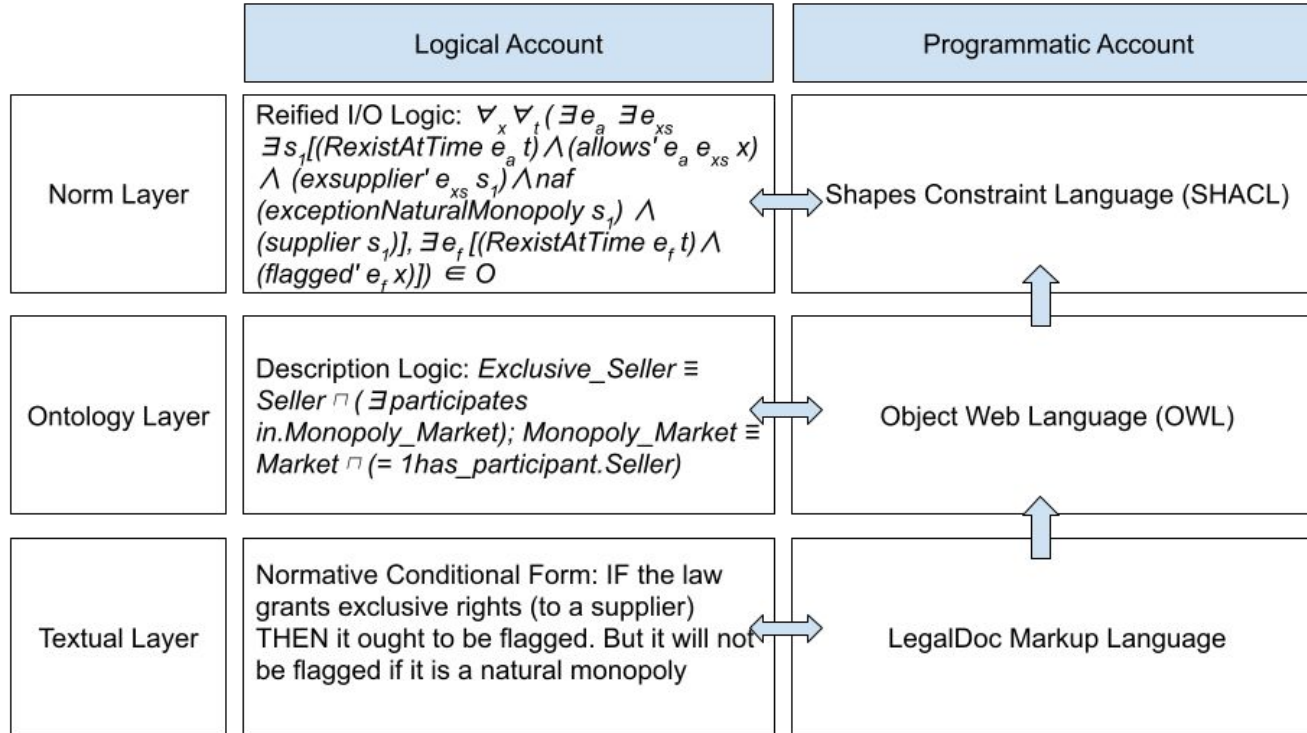
Domain to be Encoded

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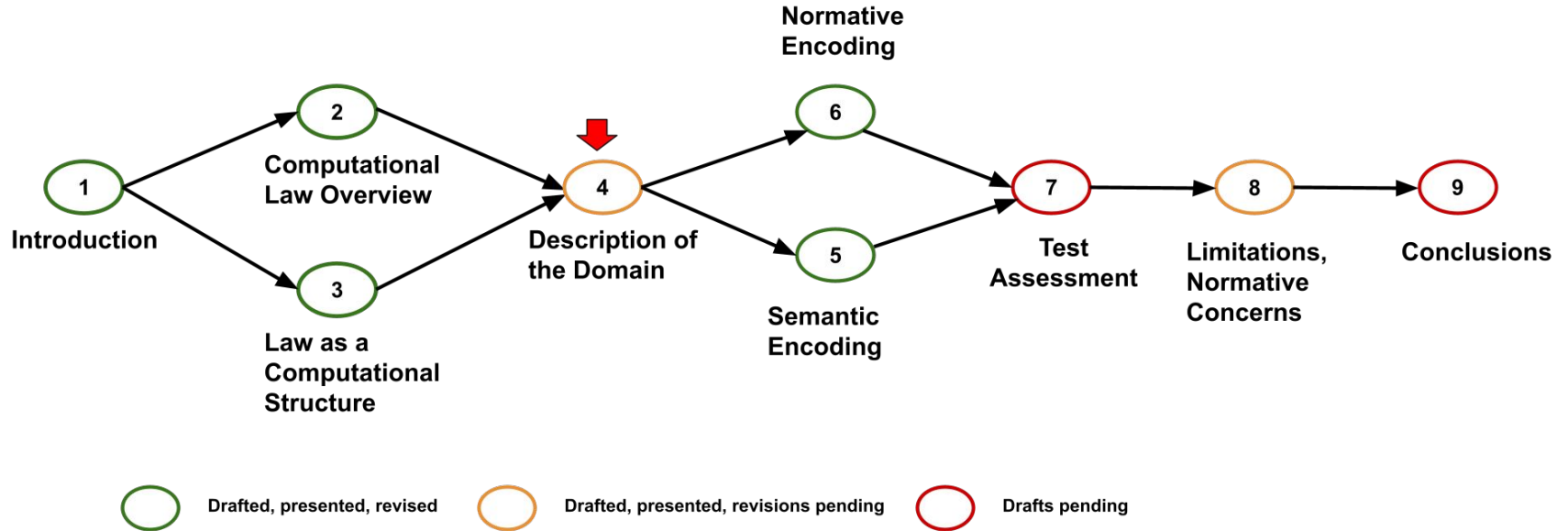
Background

Connectionist	Symbolic
Emerges from context of large training data	Explicit definitions of concepts and procedures
Statistical computations	Logical associations
LLMs, Generative AI	Rules-based systems
Control - Training data (control variables), fine tuning of model	Control - Visibility of the encoding, formalized validation and testing

Methodology



Chapter Plan

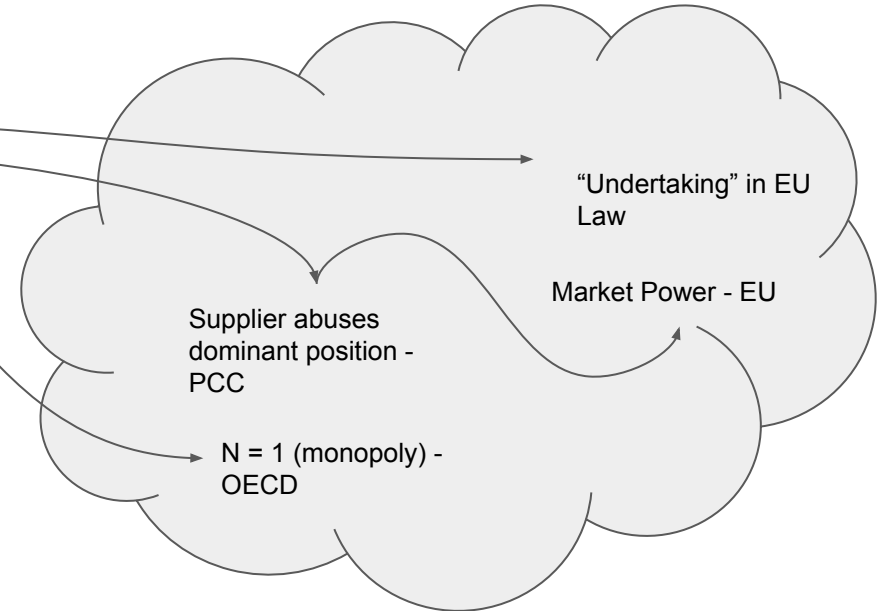


Textual Sources of the Domain

- Domain - Subject area, field of knowledge to be modeled (semantics, scoping)
 - OECD - Competition Impact Analysis (CIA) Guidelines, Workgroup on Barriers to Entry
 - EU Competition Law
 - Philippine Competition Commission - Previous CIA exercises

Code Mapping the Domain, A1

A1 - IF the law allows the supplier to offer goods and services exclusively THEN the law has to be flagged...



Unstructured text - OECD, EU Law, PCC CIA Exercises

A2 and A5 as straightforward tests

- A2 - test turns on presence or absence of prior authorization, usually reflected in a document
- A5 - whether or not prior authorization is required for movement of goods or services from one geographic area to another. States define geographic areas and jurisdictions

A3 - Limitation

- Need to model the entirety of strategic space?
 - We can assume simplified game?
- Dominance of incumbents a limitation
- Or - any significant post-entry, pre-exit costs?
 - What is significant?

A4 - Barriers to entry and exit

- Well documented debate on barriers, definition
- Established signals:
 - Sunk costs, including entry costs
 - Structural barriers
 - Strategic behavior (Dominance of incumbents)

Issues

- Resolving conceptual overlaps -
 - Exclusivity is also a barrier (OECD)
 - Abuse of dominance is a barrier (PCC)
 - A license requirement can lead to exclusivity
 - A license can also be an entry cost (barrier)

Issues

- Computational encodings may not be isomorphic with the OECD threshold tests
 - A1 - Two alternative tests
 - A2 and A5 - Isomorphic
 - A3 and A4 - A shared library of related tests for costs, certain behaviors

Next

- Programming and testing - a “toy” problem
- Test implementation with existing neural net(?)
- Drafting conclusion
- Consultations and Revisions

THANK YOU!



emersonbanez.github.io/dissertation_public